



**Name of the module:** nco\_phase\_ramp

**Target delivery date:**

**Location in GIT repository:** vspa\_common\vspa-lib

**Overview:**

This module provides kernels for performing a scaled phase ramp as follows:

$$x(k) = g * \exp\left(-j * 2 * \pi * \frac{f}{2^{32}} * (k_i + k)\right) \quad k = 0 \dots N - 1$$

**VCPU Function API:**

```
void phase_ramp_gen(cfloat16_t *out_p,  
                   cfloat32_t *gain_cpx_p,  
                   int32_t     phase_ramp,  
                   int32_t     phase_init,  
                   uint32_t     num_lines);
```

where

out\_p - pointer for the output buffer with phase ramp (complex Half Precision)

gain\_cpx\_p - pointer for complex gain (complex Single Precision)

phase\_ramp – Linear phase ramp “ $f$ ” (signed 32-bit integer, 1’s complement)

phase\_init – Initial phase “ $k_i$ ” (signed 32-bit, 1’s complement)

num\_lines – number output lines, each containing 32 complex output samples.

**DMEM requirement:**

| Variable | Minimum size<br>(half-words) | Minimum alignment<br>(half-words) | Allocated<br>by |
|----------|------------------------------|-----------------------------------|-----------------|
| out      | num_lines * 64               | 64                                | Caller          |
| gain_cpx | 4                            | 4                                 | Caller          |

**Test Plan:**

Verify the nco\_phase\_ramp output with matlab reference.